



DSA – Sec A

Lab – 05

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Part A:

1. **Fibonacci Number:** Write the recursive and iterative code for Fibonacci Number computation.
 - Analyze why Iteration is working so fast as compare to recursive implementation of Fibonacci Numbers.
 - Use **Memorization Technique** to make the recursive algorithm fast.
2. **Factorial:** Write a recursive function for computing the factorial of a number N.
3. **Permutation, Combination:** Write a recursive function using the recursive definition of *Permutation, Combination (Formula)*.
4. **TriSum:** Write the recursive and iterative code for Computing the TriSum sequence:
 - 1, 2, 3, 6, 11, 20, 37,
 - Write the recursive mathematical formulation.
 - Write the recursive code for the Sequence generator
 - Analyze what will be its time complexity (the approximate number of time the recursive call will be called).
5. **Power:** Write a recursive function to compute x raised to the power of y (i.e., x^y).
 - Function Signature: `int power(int x, int y);`
6. **Tower of Hanoi:** Write a recursive function to solve the Tower of Hanoi problem, which involves moving a stack of disks from one peg to another using a third peg, with the constraint that no disk may be placed on top of a smaller disk.

PART B:

1. **All Permutations:** Write a recursive function to generate all permutations of a given string.
Function
 - Signature: `void permute(std::string str, int l, int r);`
 - Input: `std::string str = "aab"; permute(str, 0, str.length() - 1);`
 - Output:

```
aab
aab
aba
aba
baa
baa
```
2. Fast Multiplication: Only by using '+'
 - FastMul(X, Y) should return in $O(\log Y)$
 - Help: $R=X, C=1 \rightarrow$ At each iteration $R+=R, C+=C \sim C+C \leq Y$